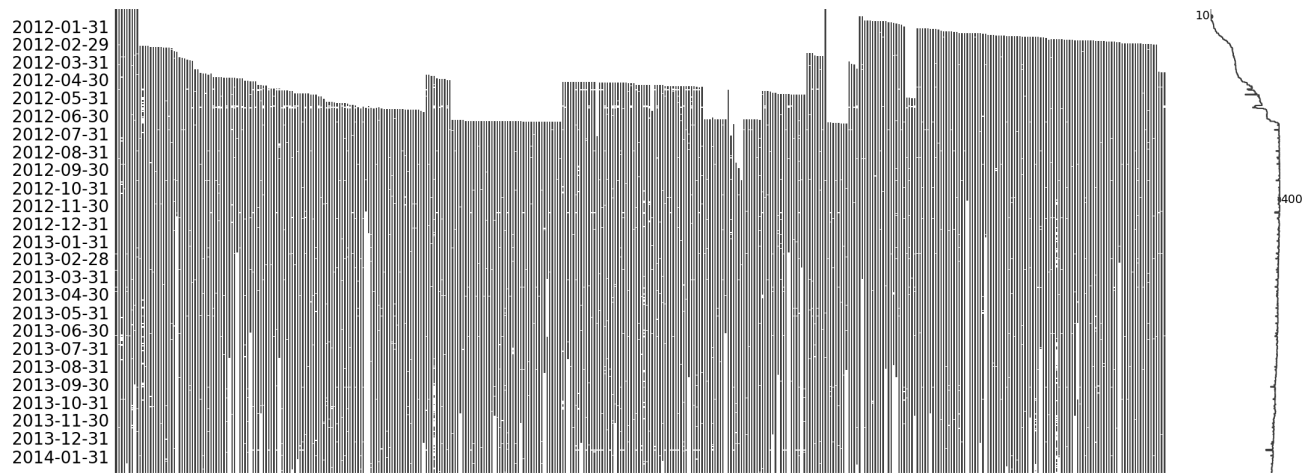


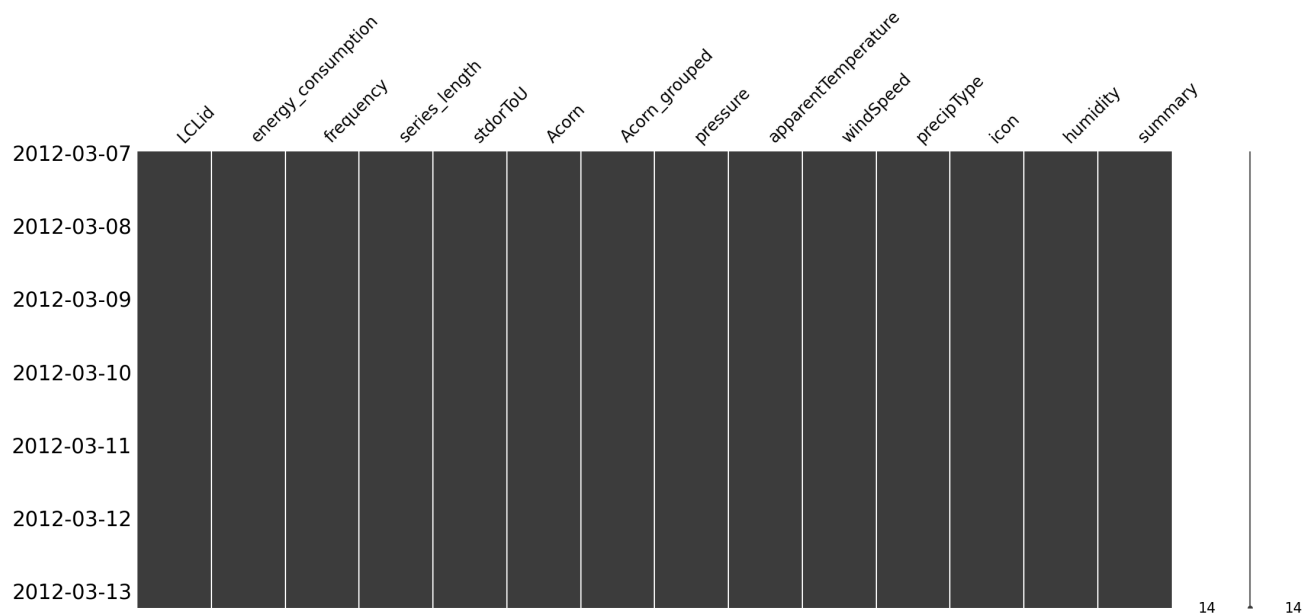
Preprocessing Time-Series Data

Missing Data and Imputation Techniques

''' Pivot data to set index as datetime & different time series in columns '''

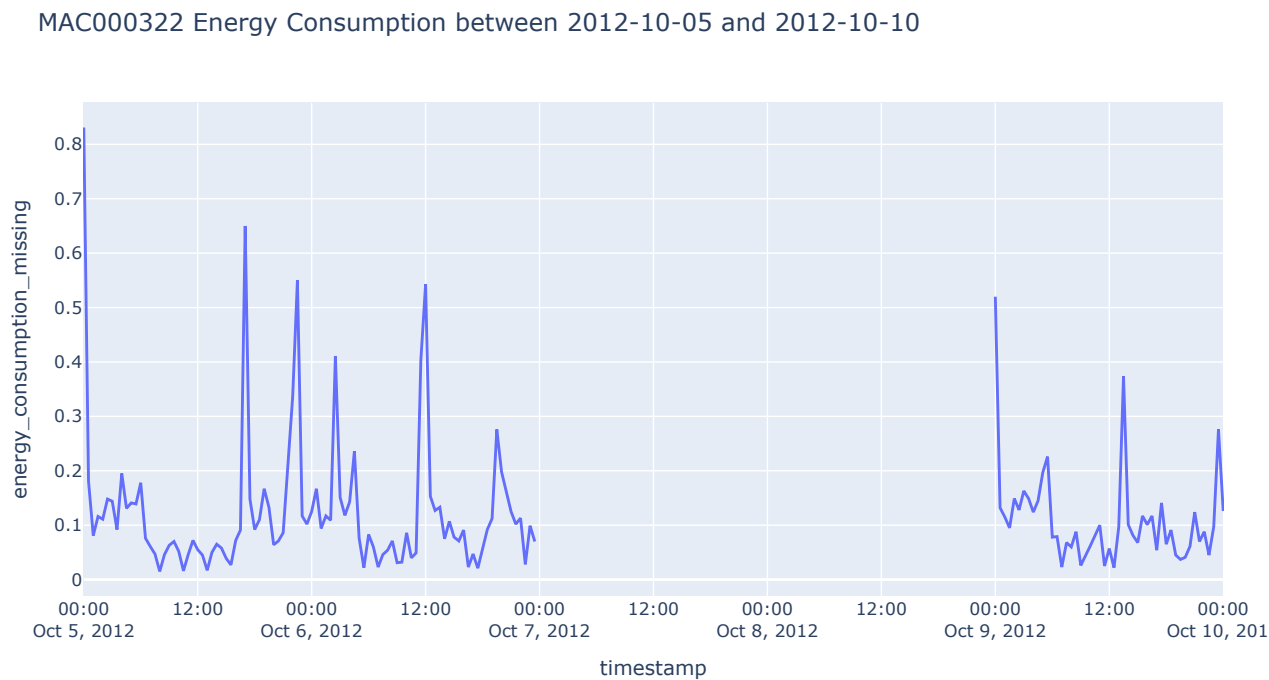


''' take a single timeseries block from the data - LCLid=="MAC000322" '''



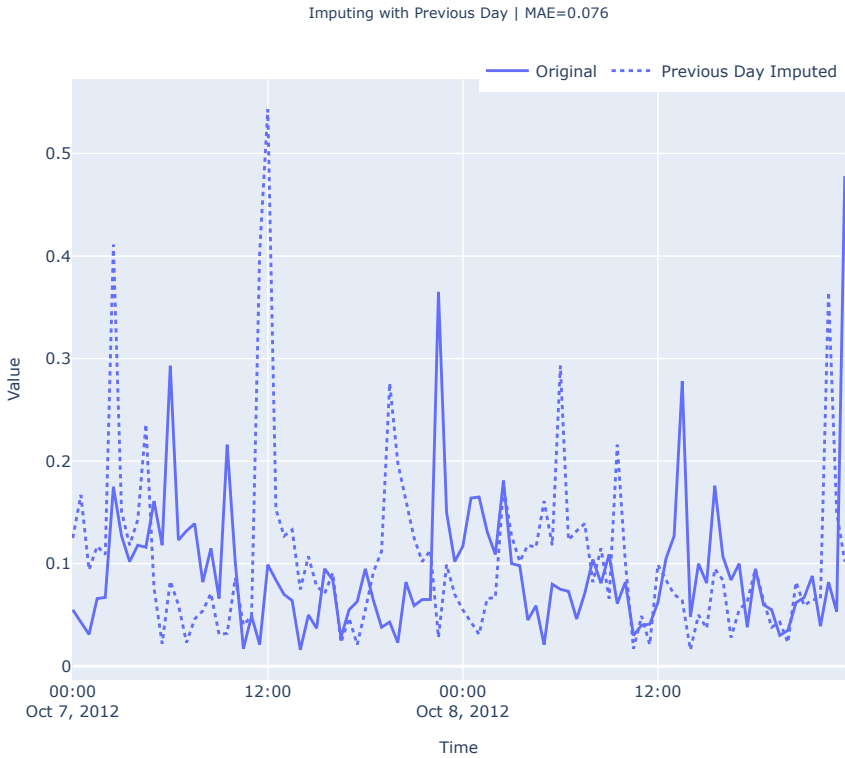
Create a missing data section to impoute the data and assess how the imputation technique works.

''' MAC000322 Energy Consumption '''



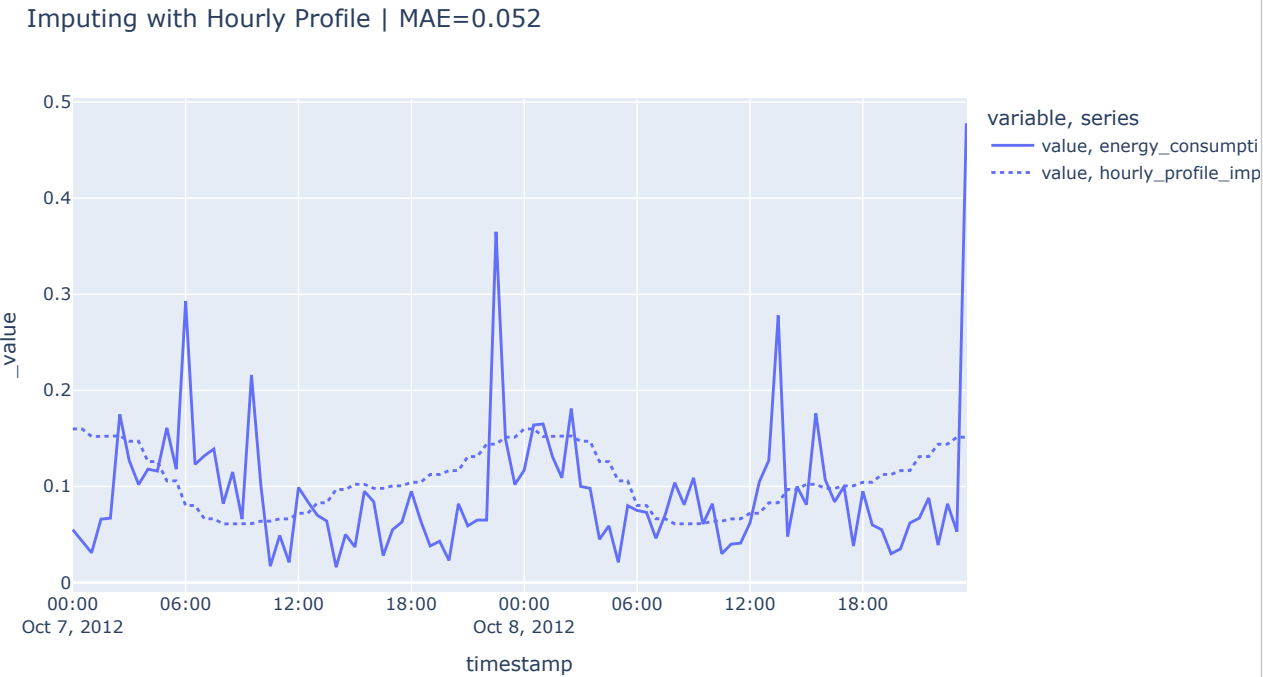
Previous day

''' Imputing with Previous Day '''



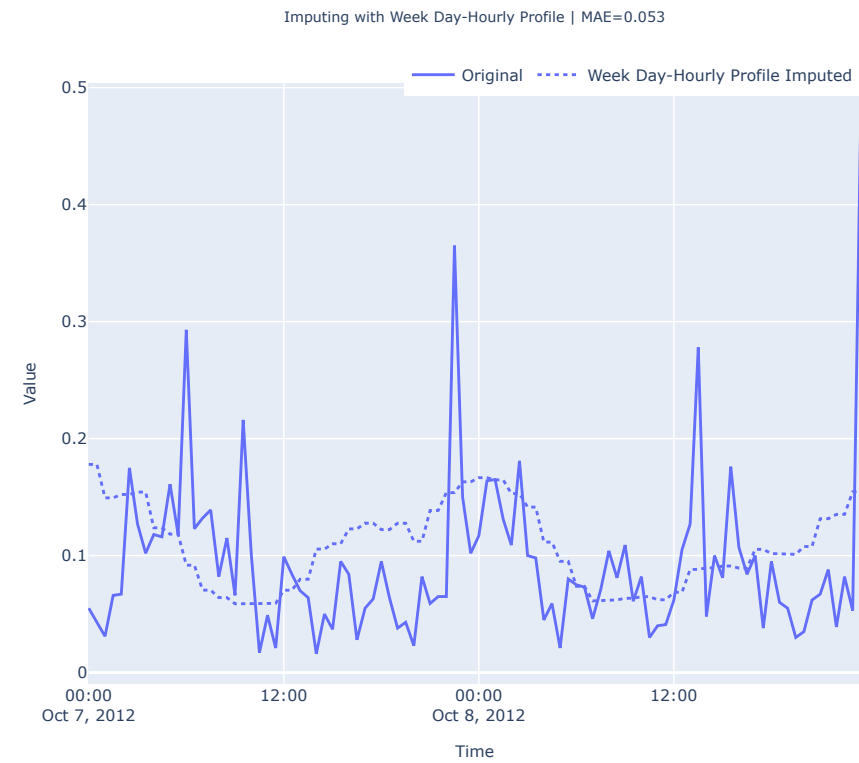
Hourly profile

''' Imputing with Hourly Profile '''

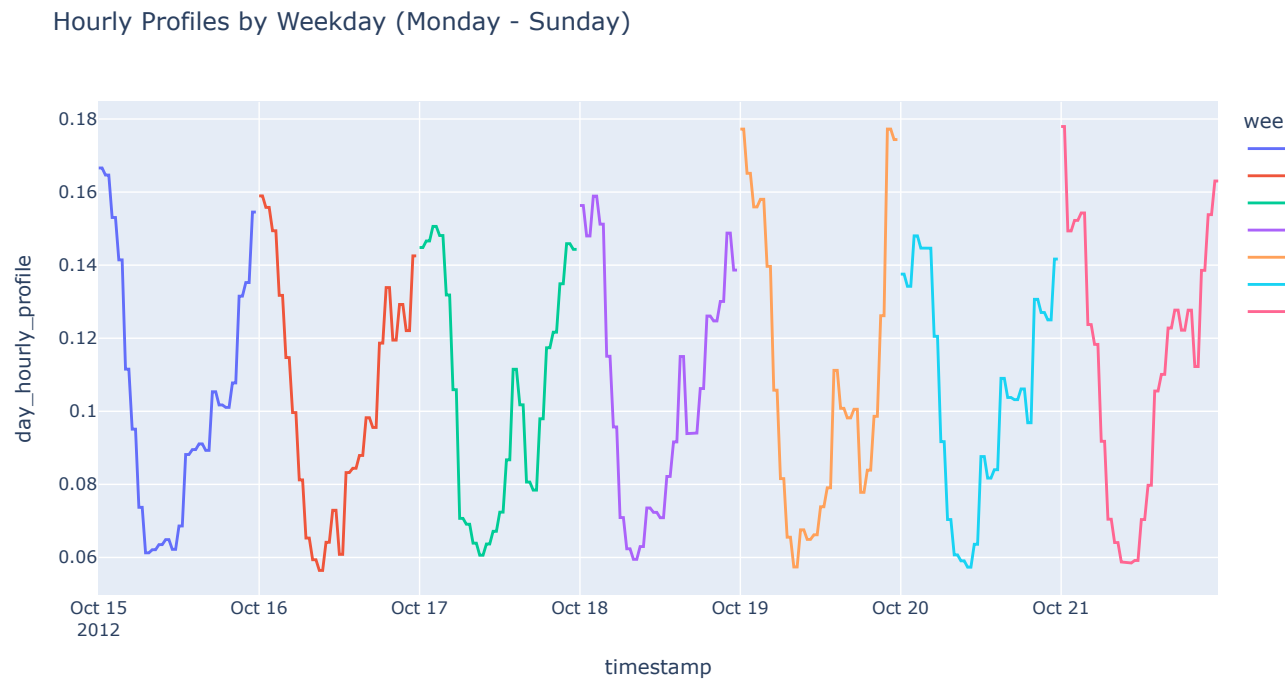


Weekly hourly profile

''' Imputing with Week Day-Hourly Profile '''



''' Hourly Profiles by Weekday (Monday – Sunday) '''



Seasonal Interpolation

''' Imputing with Seasonal Interpolation '''

Imputing with Seasonal Interpolation
MAE Half Hourly=0.054 | MAE Weekday-Half Hourly=0.055

